## 1.1

★ The 20 sentences with the highest and lowest total negative log likelihoods across the two datasets combined.

20 lowest
[en]  18.070  (tokens=5)  Here is a typical case.
[ru]  22.256  (tokens=5)  Вот характерный случай.
[en]  23.863  (tokens=7)  All the wounded were taken to hospital.
[ru]  27.991  (tokens=13)  Все раненые доставлены в больницу.
[ru]  31.344  (tokens=18)  Один из злоумышленников задержан, второму удалось скрыться.
[en]  31.668  (tokens=7)  Civil rights group issues travel warning for Missouri
[ru]  32.858  (tokens=22)  Владимир Путин поручил ликвидировать загрязнения Байкала
[en]  36.550  (tokens=12)  "We are hearing complaints at a rate we haven't heard before."
[en]  40.870  (tokens=8)  The problem was systemic throughout the entire country.
[en]  41.670  (tokens=12)  Vladimir Putin has ordered the cleanup of pollution in Baikal
[ru]  42.162  (tokens=13)  Проблема была системной в целом в стране.
[en]  43.285  (tokens=15)  One of the perpetrators has been apprehended, but the second managed to escape.
[ru]  44.380  (tokens=26)  Два человека погибли в результате стрельбы по полицейскому патрулю в Египте
[ru]  46.116  (tokens=13)  «У вас люди подвергаются нарушению прав человека.
[ru]  47.161  (tokens=14)  «Мы слышим жалобы часто, как никогда раньше».
[ru]  47.339  (tokens=29)  Проект закона был разработан Минэкономразвития России по инициативе Генпрокуратуры.
[en]  48.665  (tokens=11)  "You have violations of civil rights that are happening to people.
[en]  50.413  (tokens=15)  Two people have died as a result of a shooting of a police patrol in Egypt
[en]  51.290  (tokens=17)  It is the first such warning that the organization has issued for a state in the US.
[en]  55.151  (tokens=20)  Now, in a number of regions, they have managed to mitigate the problem to the greatest extent possib

20 highest
[ru]  164.112  (tokens=70)  В предупреждении также упоминается недавний отчёт Генеральной прокуратуры штата Миссури, согласно ко

[en] 174.598 (tokens=73) The new law will add an article to the Code of the Russian Federation on Administrative Offences, wh
[ru] 177.061 (tokens=54) Результат - нарастающий финансовый ком из долгов по зарплатам, налогам, кредитам, который привел бы
[ru] 184.287 (tokens=62) Также предупреждение по передвижению является ответом на новый закон штата Миссури, после принятия к
[ru] 187.861 (tokens=77) Сандерс погиб при подозрительных обстоятельствах ранее в этом году после того, как у него закончился
[ru] 190.354 (tokens=73) Потому что сами бизнесмены не могли справиться с недобросовестными заказчиками, которые заключали не
[ru] 191.532 (tokens=89) Согласно последним данным программы ФБР по сообщению о преступлениях на почве ненависти, в Миссури з
[ru] 192.691 (tokens=70) Предупреждения по передвижению обычно выпускаются Государственным департаментом применительно к друг
[en] 195.410 (tokens=55) “It would be right to reassure all our citizens and tourists and to make an announcement about this,
[ru] 201.292 (tokens=79) Группа сослалась на такие инциденты, как унижение чернокожих студентов по расовому признаку в Универ
[en] 217.615 (tokens=71) "The NAACP Travel Advisory for the state of Missouri, effective through August 28th, 2017, calls for
[ru] 217.696 (tokens=73) Их останавливают на дорогах из-за цвета кожи, их избивают и убивают», - рассказал газете «Канзас-Сит
[ru] 240.948 (tokens=73) "Было бы правильно успокоить всех наших граждан и туристов и заявить об этом, сказать, что это к вер
[ru] 269.991 (tokens=122) Ранее Американский союз защиты гражданских свобод (ACLU) объявил предупреждения для путешествующих п
[en] 277.467 (tokens=85) The realization of Federal Law No. 189-FZ “On amending the Code of the Russian Federation on Adminis
[en] 278.877 (tokens=99) Since 2015, 9,700 petitions, in the interests of protecting the rights of business owners to receive
[ru] 306.983 (tokens=127) С 2015 года в интересах защиты прав предпринимателей на получение оплаты по публичным заказам прокур
[ru] 328.594 (tokens=130) Качественно улучшить платежную дисциплину публичных заказчиков сможет реализация Федерального закона
[ru] 341.065 (tokens=118) По словам Национальной ассоциации содействия прогрессу цветного населения, объявить предупреждение п
[ru] 406.308 (tokens=150) Заявление группы звучит так: «Предупреждение для путешествующих Национальной ассоциации содействия п

★ The 20 sentences with the highest and lowest total negative log likelihoods, normalized by sequence length, across the two datasets combined.

20 lowest
[ru] 1.494 (tokens=22) Владимир Путин поручил ликвидировать загрязнения Байкала
[ru] 1.632 (tokens=29) Проект закона был разработан Минэкономразвития России по инициативе Генпрокуратуры.
[ru] 1.684 (tokens=59) Президент России Владимир Путин подписал закон о введении административной ответственности за наруше
[ru] 1.707 (tokens=26) Два человека погибли в результате стрельбы по полицейскому патрулю в Египте
[ru] 1.741 (tokens=18) Один из злоумышленников задержан, второму удалось скрыться.
[ru] 1.835 (tokens=32) В итоге была погашена задолженность перед предпринимателями на сумму более 66 млрд рублей.
[ru] 1.990 (tokens=45) А если должностное лицо уже подвергалось административному наказанию за аналогичное правонарушение,
[ru] 2.034 (tokens=80) Новый закон дополнит Кодекс Российской Федерации об административных правонарушениях статьей, котора
[ru] 2.054 (tokens=67) Глава Ростуризма Олег Сафонов заявил в интервью телеканалу РБК, что "положительно" относится к приня
[ru] 2.152 (tokens=89) Согласно последним данным программы ФБР по сообщению о преступлениях на почве ненависти, в Миссури з
[ru] 2.153 (tokens=13) Все раненые доставлены в больницу.
[ru] 2.157 (tokens=69) Тем не менее по состоянию на первое полугодие 2017 года совокупный объем долга по государственным и
[ru] 2.168 (tokens=40) Речь идет о том, что авиакомпании могут сами определять количество мест и вес багажа, включенных в н
[ru] 2.178 (tokens=49) Ранее президент России Владимир Путин подписал закон, предусматривающий право авиакомпаний исключать
[en] 2.211 (tokens=49) Missouri recorded 100 hate crimes in 2015, according to the latest figures from the FBI's hate crime
[ru] 2.213 (tokens=122) Ранее Американский союз защиты гражданских свобод (ACLU) объявил предупреждения для путешествующих п
[ru] 2.230 (tokens=47) Президент России Владимир Путин на совещании по вопросам защиты Байкала сообщил, что за последние го
[ru] 2.283 (tokens=31) Во время перестрелки два человека, в том числе один полицейский, погибли, три человека получили ране

[ru] 2.285 (tokens=29) Ростуризм поддерживает закон об отмене бесплатного багажа для невозвратных авиабилетов
[ru] 2.297 (tokens=44) Неизвестные обстреляли полицейский патруль в египетском городе Исна, сообщает Reuters со ссылкой на

20 highest
[en] 3.699 (tokens=25) Just two years ago, situations in which business owners were unable to collect payment for already-e
[en] 3.735 (tokens=38) A recent Missouri law making it harder for people to win discrimination lawsuits, as well as the sta
[en] 3.778 (tokens=25) The person invested all of their working capital, fully satisfied their commitments, but didn’t rece
[en] 3.827 (tokens=25) This law is meant to solve the very serious issue of customers failing to fulfill their commitments
[en] 3.837 (tokens=37) The result is an escalating financial snowball comprised of payroll, tax, and loan liabilities, whic
[en] 3.848 (tokens=37) “It is also necessary to deal with the facilities that continue to operate here; the sewage treatmen
[en] 3.864 (tokens=25) "People need to be ready, whether it's bringing bail money with them, or letting relatives know they
[en] 3.876 (tokens=26) In the regions where debt indicators continue to climb to critical levels, the prosecutor’s office h
[en] 3.890 (tokens=17) Mr. Putin also ordered that a full-fledged reclamation of polluted areas be conducted.
[en] 3.905 (tokens=24) Notably, these vast sums were by no means accrued as a result of non-payment of multibillion-ruble c
[en] 3.927 (tokens=25) All the prosecutors for the constituent entities of the Russian Federation have already received ins
[en] 3.954 (tokens=21) He instructed that sources of financing, plans, and those responsible for implementing the cleanup o
[en] 4.303 (tokens=36) The prosecutor general is personally holding on-site meetings with local officials, during which the
[en] 4.384 (tokens=37) After all, businessmen couldn’t deal on their own with dishonest customers who signed agreements wit
[en] 4.424 (tokens=11) "You have violations of civil rights that are happening to people.
[ru] 4.451 (tokens=5) Вот характерный случай.
[en] 4.524 (tokens=7) Civil rights group issues travel warning for Missouri
[en] 4.693 (tokens=19) The Russian Federal Tourism Agency supports the law eliminating the free baggage allowance for non-r

[en]  4.895  (tokens=23)  Numerous systematic delays in payment for services to business owners for considerably more modest a
[en]  5.109  (tokens=8)  The problem was systemic throughout the entire country.

★ What properties do high-likelihood sequences have that low-probability ones don’t?

High-likelihood sequences are short and use simple, common words in predictable news phrasing, such as "Here is a typical case." Low-likelihood sequences are long and full of hard-to-predict tokens: rare names and acronyms (NAACP, ACLU), numbers and dates, and legal citations like "Federal Law No. 189-FZ".

★ How does normalizing by sequence length influence this?

Normalizing removes most of the length effect, and the ranking flips by language: 19 of the 20 lowest per-token NLLs are Russian and 19 of the 20 highest are English. This is mostly a tokenization artifact, because Russian words split into several subword pieces and the later pieces are easy to predict. Short sentences also move toward the bottom, since they have little context to make their tokens predictable. Long sentences with repeated, formulaic phrases move toward the top. Per-token NLL corrects for length but can't be compared fairly across languages that tokenize differently.

## 1.2

★ Why might temperature result in the observed differences?
As temperature rises, the vocabulary spreads out sharply, from 27 distinct tokens at T=0 to 962 at T=0.7, 1,841 at T=1.0 and 3,529 at T=1.2. Dividing the logits by T before the softmax makes the distribution more peaked when $T < 1$ and flatter when $T > 1$. At T=0 sampling becomes greedy, so all 50 samples are the same repetitive loop, made of a few tokens like print. At T=0.7 and T=1.0, more probability goes to less likely tokens, so the text becomes more varied and moves from code to ordinary English. At T=1.2, the tail of the distribution gets so much probability that output becomes incoherent.

★ Why might ϵ result in the observed differences?
Raising ε steadily shrinks the vocabulary, from 1,997 distinct tokens at ε=0 to 942 at ε=0.05 and 503 at ε=0.1. ε-sampling drops every token whose probability is below ε and renormalizes over what's left, so the low-probability tail is never sampled. With ε=0 nothing is filtered and the samples are ordinary T=1 samples: varied, spread-out text across many topics. At ε=0.05 and ε=0.1, generation is limited to the model's few most confident choices at each step, so it keeps falling back to the most likely continuation, a

Python "hello" string example. As a result, print, (h and ello dominate the counts, and repetition increases. ε trades diversity for coherence like lowering the temperature does, but it removes the tail entirely instead of just shrinking its probabilities.

★ Why might k result in the observed differences?
Top-k sampling keeps only the k most likely tokens at each step and renormalizes over them, so k caps how many different tokens can be picked. With k=1, only the single most likely token is ever chosen, which is greedy decoding. The output matches T=0 exactly. Raising k to 20 brings in real variety, with 1,350 distinct tokens and text that switches between code, English and Chinese. Going from 20 to 100 adds comparatively little, giving 1,595 distinct tokens, still fewer than unfiltered T=1 sampling at about 1,850 to 2,000. That's because most of the probability usually sits in the top 20 tokens, so the extra 80 are rarely sampled. Unlike ε or top-p, k is fixed no matter how confident the model is. It can cut off reasonable options when the distribution is flat, and it can let in unlikely ones when the distribution is peaked.

★ Why might p result in the observed differences?
Top-p sampling keeps the smallest set of top tokens whose combined probability reaches p, then renormalizes. A larger p therefore keeps more of the tail, and the vocabulary grows: 994 distinct tokens at p=0.5, 1,528 at p=0.9 and 1,864 at p=1.0. At p=1.0 nothing is filtered, so it's the same as plain T=1 sampling. With the same random seed, it even produces the same first sample as the T=1 run. At p=0.5, the model sticks to its most confident continuations, so the "hello" Python examples (print, (h, ello) dominate the counts. Larger p allows more varied English and Chinese text. Unlike top-k, the number of kept tokens changes from step to step: it's small when the model is confident and large when it's unsure. Lengths barely change, with a mean of 85 to 90 tokens before the first EOS for every p. Every sample is exactly 100 tokens by design, and p mainly affects which tokens are chosen, not when an EOS appears.

## 2.1

bigram validation perplexity: 1448.8367175110927

## 2.3

trigram backoff validation perplexity: 678.4869796067352
trigram Kneser-Ney backoff validation perplexity: 620.477474319148

## 2.4

Neural trigram validation perplexity: 541.04

## 2.5

LSTM validation perplexity: 318.38

## 3.1

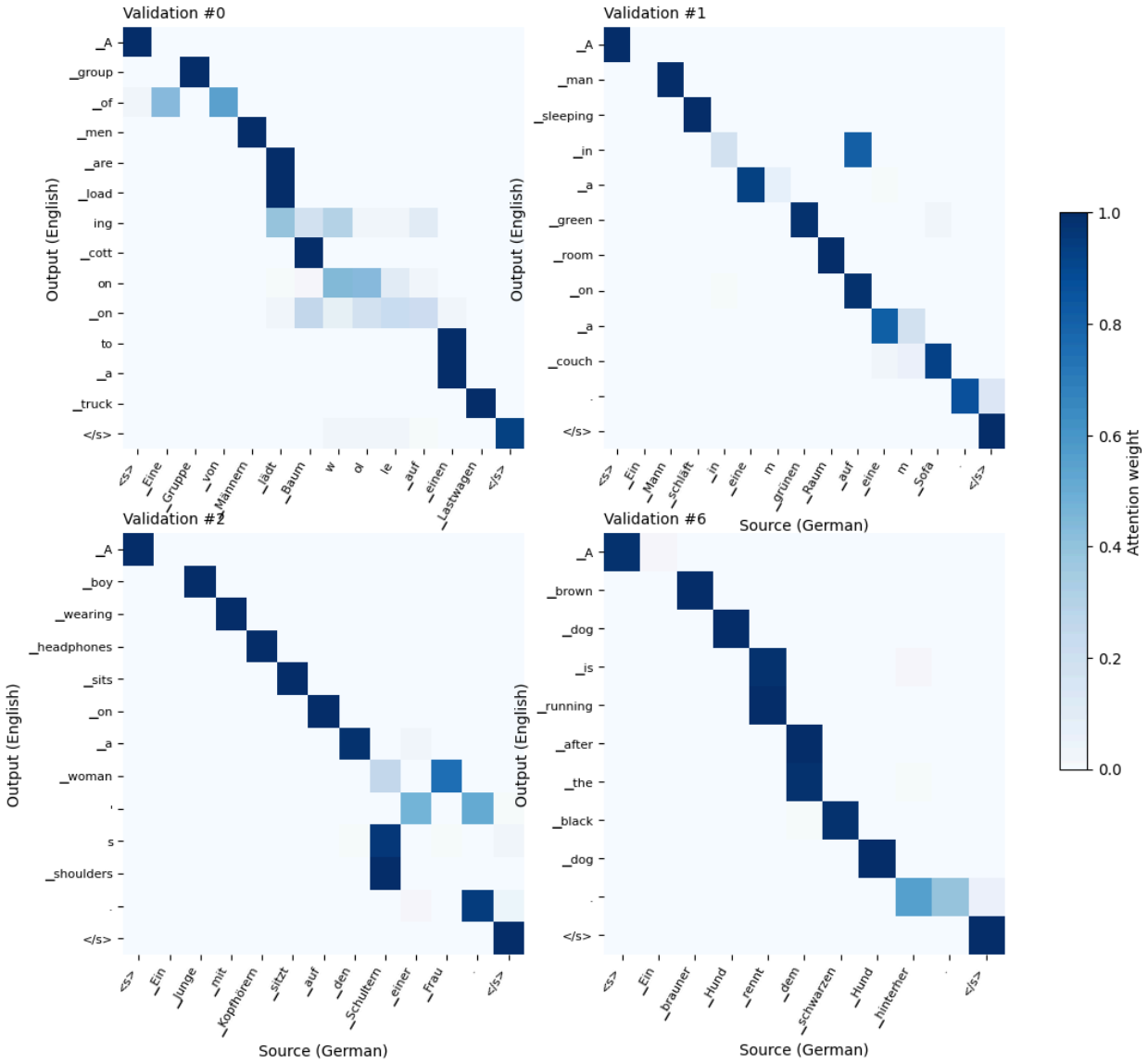
★ Report the performance of each method using accuracy, precision, recall, and F1 score. Which method performs the best? Which performs the worst? Why?

mean: acc 0.6927  P 0.6919  R 0.6948  F1 0.6933
last: acc 0.6133  P 0.6005  R 0.6771  F1 0.6365

The mean-embedding classifier performs best on every metric, and the last-token classifier performs worst. Averaging the LSTM's final-layer outputs over every token collects sentiment cues from the whole review, such as "boring" in one sentence and "brilliant" in another. A single hidden state at the last token mostly reflects the last few words. The LSTM was trained to predict the next word, not to carry a review-level judgment across hundreds of tokens, so earlier sentiment fades from its state. Reviews are also cut off at 256 words, so for long reviews the "last" token is somewhere in the middle, not the concluding verdict.

## 4.2

★ A figure with attention map plots for 4 sentence pairs from the validation set. We encourage you to look through more maps to aid your analysis, but please only include 4 representative plots in the figure.




★ A brief discussion over trends you discover in the plots. Do the maps line up with your intuition, are there any surprising alignments? Are there any many-to-one or many-to-many alignments, or mainly one-to-one? Using a tool like Google Translate on substrings may help give some insight into this.

Most of the attention lies close to a diagonal, and most alignments are one-to-one, as I expect. Nouns and adjectives line up cleanly with their translations: Mann→man, grünen→green, Raum→room, etc. However, many-to-one and many-to-many alignments appear when words are split differently, such as "are loading" aligning with "lädt" and "cotton" with "Baumwolle." Some word reordering also creates crossing alignments, such as "a woman's shoulders."

Surprisingly, some words attend to context rather than their direct translations. For example, "after" attends to "dem" instead of "hinterher." Overall, attention generally reflects word alignment but does not always correspond directly to translations.